

Verifier-Guided Evaluation of LLM-Generated Network Configurations: a Paired Study with Shared Initial Candidates

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Abstract—We preregistered and executed a paired confirmatory study (CAND-2026-001) that measured whether a verifier-triggered guarded pipeline (generate $\rightarrow$ validate $\rightarrow$ repair, ≤ 1 repair) reduces deterministic-invariant violations relative to single-shot initial generation for intent $\rightarrow$ configuration NetOps tasks. Using a deterministic synthetic task generator and a deterministic network verifier, we ran 200 preregistered tasks under shared_initial_candidate semantics with the hosted model gpt-5-mini (execution/model_configuration.json records temperature = null/unset). Baseline configurations passed the verifier in 199/200 tasks (99.5%); the guarded pipeline passed 200/200 (100%). Paired absolute risk difference (guarded - baseline) = 0.005; contingency counts n11=199, n10=1, n01=0, n00=0; exact two-sided McNemar $p = 1.0$; paired-bootstrap 95% percentile CI = [0.00, 0.015] (10,000 resamples, seed 1729). The single guarded improvement arose from one verifier-triggered repair that corrected a multi-step ordering violation. We report preregistration, full execution accounting (201 model calls: 200 shared initial + 1 repair), paired analysis, representative diagnostic artifacts, and bounded limitations grounded in the archived execution bundle.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate high-level operator intent into device configurations for network operations (NetOps). Operational correctness for such workflows requires generated configurations to satisfy deterministic invariants (reachability, policy compliance, workflow ordering, and group-level safety). Prior prototype systems and benchmarks illustrate feasibility but leave open questions about how much deterministic verification and bounded repair alter acceptance rates when the identical initial sample is reused across comparison conditions.

This paper reports a preregistered paired experiment (CAND-2026-001) designed to isolate the marginal effect of a verifier-triggered automated repair on an identical generated candidate. Under shared_initial_candidate semantics the same single initial generation is deterministically reused by both baseline and guarded episodes; any paired difference therefore arises solely from the permitted validator-triggered repair. The primary estimand is the paired difference in per-task binary acceptance by a deterministic verifier (paired_success_rate_difference_guarded_minus_baseline). Because shared_initial_candidate semantics were enforced, this estimand measures a repair-only effect and does not capture potential benefits from independent guarded first-pass generations, constrained prompts, or multi-sample selection.

Contributions. (1) A preregistered, fully archived paired experiment executing 200 intent $\rightarrow$ configuration tasks with a deterministic verifier and a hosted LLM (gpt-5-mini) under shared_initial_candidate semantics. (2) Complete execution accounting and deterministic reconciliation showing 200 shared initial generations and 1 repair call (201 model calls total). (3) Artifact-grounded confirmatory analysis reporting baseline pass 199/200 and guarded pass 200/200 with contingency counts (n11=199, n10=1, n01=0, n00=0), paired risk difference 0.005, exact two-sided McNemar $p = 1.0$, and bootstrap 95% CI = [0.00, 0.015] (10,000 resamples, seed 1729). (4) A localized failure diagnosis for the single repaired pair using archived validator traces. (5) Detailed reproducibility guidance and bounded limitations tied to archived artifacts and reconciliation records.

II. RELATED WORK

We conducted a fresh literature investigation and verified records covering intent $\rightarrow$ configuration systems, generate-validate-repair patterns, and agentic-AI safety discussions relevant to NetOps. Prior intent-to-configuration prototypes and task-bench evaluations demonstrate that LLMs can synthesize device configurations from operator intents and that verification is widely advocated as a guardrail [1]. Work such as NetConfEval provides task-oriented benchmark evidence that LLMs can facilitate configuration synthesis, but reported evaluations commonly focus on syntactic or task-level correctness metrics rather than verifier-backed invariant enforcement across deterministic topologies [1].

Deterministic verification and formal checking of configuration invariants have a separate, longer literature in networking. Beckett et al. present a general approach to network configuration verification that emphasizes encoding device semantics and network-wide properties into analyzable specifications; such verifier-focused methods clarify the kinds of invariants a downstream oracle must implement to make generator $\rightarrow$ validator pipelines meaningful in practice [2]. Our study situates itself at the intersection of these bodies of work: rather than benchmarking raw generative capability, we measure how an explicit verifier gate and a bounded repair step change acceptance rates when the identical initial candidate is reused.

Cross-domain empirical work on generate $\rightarrow$ validate $\rightarrow$ repair and test-in-the-loop LLM repair

in software engineering provides a methodological precedent. Test-driven or test-in-loop program-repair demonstrations show that pairing generation with a deterministic test oracle (failing test $\rightarrow$ patch $\rightarrow$ regression test) yields reproducible corrections for many classes of faults, and those studies emphasize artifact archiving to permit independent reproduction [4]. Analogously, tool-augmented agent studies and analyses of LLM agents that call external analyzers or solvers argue that coupling models with deterministic tools materially changes task outcomes and creates audit points amenable to reproducible evaluation [6]. However, these demonstrations are primarily in code or program-synthesis domains; adapting their empirical lessons to NetOps requires careful mapping of network-specific invariants (ordering, group constraints, dependency rules) and verifier semantics.

Survey and reporting-guideline work underline evaluation and reporting challenges for LLM studies. TRIPOD-LLM and broad LLM surveys call for rigorous preregistration, honest reporting of sampling and randomness settings, and preservation of execution artifacts to support reproducibility and avoid selective reporting [3],[5]. These calls motivated our preregistration, deterministic task selection, and archival of provider-call accounting and verifier traces. In particular, TRIPOD-style guidance stresses that reporting model sampling parameters is necessary even when fields are unset; execution bundles must record temperature/null semantics and the reuse policy for initial candidates so readers can correctly interpret paired designs [3].

Comparative synthesis. Across the verified records there is a consistent pattern: (a) promising LLM-based intent $\rightarrow$ config prototypes and benchmarks exist, (b) generate $\rightarrow$ validate $\rightarrow$ repair architectures are advocated and empirically effective in adjacent domains, and (c) end-to-end verifier-integrated NetOps experiments with full artifact grounding and preregistered paired estimands are scarce. Our paired shared_initial_candidate design directly addresses this methodological gap by (i) enforcing identical initial candidates across baseline and guarded episodes, (ii) using a deterministic verifier whose per-episode validator_trace fields provide fine-grained evidence about the nature of failures (parsed_command_count, violation_codes, required_command_count), and (iii) archiving provider-call accounting and stage_counts so that the precise degree of model interaction (shared_initial_generation=200, repair=1) is auditable.

Limitations in the prior literature that motivate our approach. Many prior works evaluate generation quality via held-out references or human judgement rather than deterministic invariant enforcement; others integrate emulators or dataplane tests but do not archive the exact paired generation semantics needed to isolate repair-only effects. By contrast, the verifier-centric studies and test-in-the-loop program-repair literature demonstrate that a deterministic oracle can make repair effects measurable and reproducible, but these prior empirical results must be adapted to NetOps because network invariants include ordering and group-level constraints that differ from unit tests

used in program repair. Our contribution is therefore methodological and empirical: we apply a preregistered, verifier-grounded paired protocol to quantify the marginal repair effect under shared_initial_candidate semantics and provide the full artifact bundle needed to reproduce that narrowly scoped estimand.

III. METHODOLOGY

Preregistration and design freeze. The study design and analysis plan were frozen in preregistration CAND-2026-001 prior to any model calls. The preregistration fixed: (a) a paired binary estimand (paired_success_rate_difference_guarded_minus_baseline) comparing per-task binary verifier acceptance under baseline (no repair) versus guarded (verifier may trigger ≤ 1 repair) conditions; (b) deterministic task selection for $N=200$ tasks; (c) generation_semantics = shared_initial_candidate (one sampled initial generation per task deterministically reused by both episodes); (d) a maximum of one automated repair call per task; (e) primary failed-call treatment complete_pair_primary and a sensitivity treatment that counts persistent unparsables as failures; and (f) the confirmatory test (exact two-sided McNemar via exact binomial tail) plus a paired-bootstrap procedure for a percentile CI. The preregistration artifact (immutable) is referenced in the execution manifest.

Deterministic task sampling. Tasks were produced by a deterministic task generator (deterministic_netops_task_generator_v5) and selected by a fixed index rule documented in the preregistration. The selection rule is deterministic across the 200-task bank (the preregistration specifies the cycle-and-offset index mapping used to pick each task index); this ensures reproducible task composition and eliminates post-hoc selection. Each manifest encodes: initial_state, expected_final_state, required_sequence (ordered command list), allowed_touched_fields, workflow_policies (group-level constraints, minimum-active rules), and difficulty metadata (changed_interface_count, dependency_rule_count, required_command_count). Difficulty labels used in the study include very_hard and extreme; representative required_command_count values ranged from 3 up to 9 in the archived sample.

Shared_initial_candidate semantics and sampling notes. The adapter contract enforced shared_initial_candidate semantics: for each pair the exact same initial generation (a single sampled candidate) was provided to both the baseline and guarded episodes. Consequently, any observed paired difference can only arise from the action of the deterministic verifier and the single permitted repair; it cannot arise from independent first-pass guarded generations, different sampling calls, prompt-engineering differences, or selection among multiple generated candidates. The model-configuration record declares the requested model as gpt-5-mini and records temperature = null (unset). We explicitly note that temperature = null/unset in the configuration file is not evidence of deterministic sampling and does not imply temperature = 0; nondeterministic hosted-model sampling therefore remains a possible background prop-

erty even though the experiment used a single initial sample per pair by design.

Generation, sanitizer, and repair policy. For each task the pipeline executed a deterministic formatting sanitizer (`automatic_syntax_sanitizer_v1`) to correct minor syntactic normalization issues before verifier parsing; this sanitizer is deterministic and is applied prior to scoring per the preregistration. After the shared initial generation and sanitizer, the deterministic verifier parsed the candidate. If the verifier reported any invariant violations, the guarded logic was permitted to invoke at most one automated repair call to the hosted model with a repair prompt constructed from the validator trace (this repair policy and prompt format were specified in the preregistration). Repair calls were counted against the `maximum_model_calls` budget; the execution manifest reports exactly one repair call across all pairs in the run. The pipeline’s missingness plan treated persistent unparsable outputs (after sanitizer and an allowed repair attempt) as scorable failures in the sensitivity analysis; in the primary analysis pairs with complete episodes remained eligible per the preregistered `complete_pair_primary` rule.

Deterministic verifier as the oracle. The deterministic verifier (`deterministic_netops_validator_v5`) is the study’s operational oracle. The verifier implements: syntactic parsing into discrete commands; enforcement of `allowed_fields` and `allowed_touched_fields` per task manifest; comparison of parsed command sequences against the manifest’s `required_sequence` ordering; evaluation of group-level invariants (e.g., `final_group_constraints` and `minimum_active_in_groups`); and a final boolean `full_intent_constraint_validation = true` only if all encoded constraints are satisfied. The verifier emits structured tracer output for each episode that includes `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `normalized_configuration`, `validation_before` (violation codes and counts), `validation_after` (post-repair state and validity flag), `validator_id` and version, and a human-readable description of detected invariant breaches. These tracer fields are the authoritative evidence for per-episode scoring in the analysis.

Paired execution bookkeeping and provider accounting. The preregistered planned episodes were 400 (200 pairs $\times$ 2 conditions). The run completed all planned episodes without terminal provider failures. Provider accounting recorded hosted-model-call events and stage counts consistent with preregistration: the stage counts indicated 200 shared initial generation events and 1 repair event, for a total of 201 model calls. Cache and reuse checks verified that the shared initial candidate was reused per pair and that no cross-condition cache-reuse introduced additional independent generation calls. All runtime provider metadata (call counts, cache hits/misses, stage counts) were reconciled deterministically in the execution reconciliation artifact.

Statistical procedures — confirmatory and CI computation. The primary confirmatory test is the exact two-sided McNemar test computed from discordant pair counts `n10` and `n01` (where `n10` = count of pairs with `guarded=1` & `baseline=0`, and `n01`

= count with `guarded=0` & `baseline=1`). We implemented the exact two-sided McNemar p-value using the exact binomial-tail formulation: let $n = n10 + n01$ and $k = n10$; the two-sided p-value is $p = 2 * \min\{ \text{BinomCDF}(k; n, 0.5), 1 - \text{BinomCDF}(k - 1; n, 0.5) \}$ with the conventional handling when $k = 0$. This exact-binomial realization is documented in the analysis log and reproduces the $p = 1.0$ reported for the observed discordants ($n = 1, k = 1$). For the paired absolute risk difference we computed a paired-bootstrap 95% percentile confidence interval with 10,000 resamples and `bootstrap_seed = 1729`; bootstrap parameters and seed are recorded in the analysis log.

Missingness, exclusions, and contamination controls. The preregistration defined deterministic exclusion criteria prior to model calls (e.g., token-normalized identity to public vendor-source templates to avoid leakage) and required logging of any excluded tasks. The run recorded no preregistered exclusions. The sanitizer and missingness policy were applied deterministically; persistent unparsables did not occur in this execution. Contamination checks included token-level similarity scans between generated candidates and vendor-template sources and a contamination summary that would flag unusually high overlaps; no contamination flags were raised in this run.

Reproducibility and artifact grounding. All implementation details required to reproduce the methodology are archived in the execution bundle: the immutable preregistration, the deterministic task generator artifacts and manifests (task bank with `required_sequence` and `workflow_policies`), the model-configuration record (including declared model and the explicit temperature = null/unset entry), the verifier implementation and structured tracer outputs, provider-call accounting and deterministic reconciliation, the sanitizer and repair-prompt templates, and the analysis-executor configuration (exact McNemar code path and bootstrap seed/resample settings). The verifier’s tracer fields provide the operational link between generated text and the binary scoring used in the confirmatory analysis. Re-running the archived paired analysis executor on the raw results reproduces the contingency counts and CI parameters reported in the paper.

IV. RESULTS

Execution accounting and provider audit. The archived execution manifest records start and end timestamps (UTC) showing the run elapsed ≈ 29 minutes (start 2026-09-01T16:03:05.491039+00:00, end 2026-09-01T16:32:13.298536+00:00). `Planned_episode_count = 400` and `completed_episode_count = 400`; `failed_episode_count = 0`. Provider-call accounting (`deterministic_reconciliation.json`) reports `hosted_model_call_event_count = 201` with `stage_counts = {shared_initial_generation: 200, repair: 1}`. `Cache_miss_count = 201` and `cache_hit_count = 0` indicate fresh model generations for all shared-initial calls and a single repair-stage call; these counts match the preregistered `shared_initial_candidate` semantics and the recorded single repair invocation.

Primary confirmatory outcomes. analysis/condition_summary.csv shows baseline successes = 199 and failures = 1 (baseline success_rate = 0.995) and guarded successes = 200 and failures = 0 (guarded success_rate = 1.000). The paired contingency table (analysis/paired_contingency_table.csv) records $n_{11} = 199$, $n_{10} = 1$, $n_{01} = 0$, $n_{00} = 0$. From these counts the paired absolute risk difference (guarded - baseline) = 0.005. Using the exact two-sided McNemar formulation described in Methods, with $n = n_{10} + n_{01} = 1$ and $k = n_{10} = 1$ we compute $\text{BinomCDF}(1;1,0.5) = 1$ and $1 - \text{BinomCDF}(0;1,0.5) = 0.5$; hence $p = 2 \times \min(1, 0.5) = 1.0$. The paired-bootstrap percentile 95% confidence interval for the absolute risk difference was computed with 10,000 resamples and bootstrap_seed = 1729 (analysis/analysis_log.jsonl) and equals [0.00, 0.015]. These numeric artifacts (analysis/condition_summary.csv, analysis/paired_contingency_table.csv, analysis/analysis_log.jsonl) are archived and reproduce the reported confirmatory values when the paired_binary_analysis_v1 executor is re-run on the archived input_results_sha256.

Single repair event and diagnostic traces. The guarded pipeline invoked exactly one automated repair. Archived validator traces for the repaired pair indicate an extreme composite task (difficulty metadata recorded in the task manifest: required_command_count = 9, changed_interface_count = 3, difficulty.level = "extreme") whose shared initial candidate violated the task's required_sequence ordering constraint. The verifier trace for that episode documents validation_before.violation_codes showing the ordering breach, the shared_initial_candidate (the identical candidate reused by baseline and guarded), and validation_after with validation_after.valid = true once the bounded repair was applied. The repair prompt/trace and the validator's normalized_configuration show the localized reordering the repair enacted; after repair the scorer returned score = 1 (scoring_reason_code = "VALID_CONFIGURATION"). These per-episode fields are present in the execution/scoring/* artifacts and constitute the archived evidence that the single observed paired improvement resulted from a verifier-triggered reordering repair that converted a sequenced-dependency failure into a valid configuration under the verifier's invariants.

Missingness, exclusions, and contamination checks. The run recorded no preregistered exclusions and no persistent unparsables; analysis/missingness_summary.csv reports complete_pairs = 200 and zeros for all other missingness categories. analysis/contamination_summary.csv is empty (no flagged contamination). Deterministic_reconciliation.json indicates pair_accounting_consistent = true and all_deterministic_consistency_checks_passed = true. No episodes have terminal_error_type set, and provider-call audit details align with preregistration (200 shared initial generations, 1 repair). These artifacts support the statement that the confirmatory analysis included the full preregistered sample without post-hoc exclusions.

Exploratory diagnostics by difficulty metadata. Representa-

tive task manifests archived in execution/task_manifest.jsonl encode difficulty metadata (changed_interface_count, dependency_rule_count, required_command_count). In the archived representative subset required_command_count values span 3–9 and changed_interface_count values up to 3; the lone baseline failure is associated with the extreme composite difficulty pattern (highest required_command_count in the sample), consistent with the preregistered exploratory hypothesis that multi-device sequencing and dependency complexity concentrate residual failures. analysis/paired_difference_figure.svg (archived) visualizes condition rates and the bootstrap distribution used for the CI; those visual diagnostics and the per-task validator_trace fields (parsed_command_count, observed_touched_field_count, violation_codes) are available for targeted reanalysis to inspect whether other high-difficulty tasks produced near-miss traces that would benefit from alternative repair strategies.

Synthesis of quantitative uncertainty. The realized baseline success rate ($199/200 = 99.5\%$) places the confirmatory test in a sparse-discordant regime: with only one discordant pair the exact McNemar p-value gives limited directional evidence ($p = 1.0$), while the paired-bootstrap CI [0.00, 0.015] quantifies the upper bound of plausible absolute improvement attributable to repair in this repair-only estimand. In practical terms, these results show that, for this synthetic verifier-backed task bank and the single-model run, bounded verifier-triggered repair corrected one expressible sequencing fault; they do not imply large marginal gains across broader settings without further experiments (e.g., independent guarded-generation contrasts, multi-model studies, or live dataplane emulation).

V. DISCUSSION

We expand the interpretive analysis and diagnostics here while preserving the paper's existing supported content.

Estimand interpretation and scope (explicit). Because the experiment used shared_initial_candidate semantics, baseline and guarded episodes for each task began from the identical sampled candidate; consequently the primary estimand intentionally measures the marginal effect of permitting an automated verifier-triggered repair on that identical candidate (a repair-only effect). This design isolates repair mechanics from any potential benefits that could arise if the guarded condition were allowed to sample independently or draw multiple candidates.

Sparse-discordant regime and inferential consequences. The observed data ($n_{11}=199$, $n_{10}=1$, $n_{01}=0$) place the analysis in a sparse-discordant regime, a setting where conventional asymptotic approximations are unreliable and exact discrete tests or resampling are appropriate. Our use of the exact binomial-tail McNemar calculation and a paired-bootstrap percentile CI (10,000 resamples, seed 1729 as archived in analysis/analysis_log.jsonl) is consistent with best practice for small discordant counts. Practically, the realized baseline success rate (99.5%) substantially reduced the experiment's sensitivity to detect modest repair-only effects relative to our preregistered power target (designed for mid-range baseline

rates ~45–65%). The bootstrap CI [0.00, 0.015] quantifies that the data are compatible with effects up to 1.5 percentage points in absolute terms under our paired repair-only estimand; the preregistered design could not reliably detect larger effects because none were present.

Mechanistic diagnostic of the repaired episode. The single repair call corrected a concrete sequencing violation expressible in the task manifest’s `required_sequence` field (`required_command_count` = 9 in the repaired manifest). Archived `validator_trace` fields show `validation_before.violation_codes` listing the ordering breach, `parsed_command_count` equal to the presented commands, and `validation_after.valid` = true after the automated reordering repair. These per-episode traces demonstrate: (1) the verifier’s capacity to detect ordering invariants encoded in the manifest; (2) the repair module’s ability to perform localized, semantics-preserving reorderings that the verifier accepts; and (3) that such faults concentrated in tasks with higher `required_command_count` and extreme difficulty labels in our deterministic task bank. All diagnostic fields cited above are available in `execution/scoring/task-00000X-*.json` in the archived bundle.

Construct validity of the deterministic verifier. `deterministic_netops_validator_v5` enforces a structured subset of operational invariants encoded in the task payloads (ordering, `allowed_touched_fields`, group-level up/down constraints, and explicit per-field prerequisites). This verifier therefore provides a high-fidelity oracle for manifest-encoded invariants but does not emulate dataplane behavior, vendor runtime idiosyncrasies, or external control-plane side-effects. Construct-valid conclusions should therefore be framed strictly in terms of manifest-specified, deterministically verifiable invariants; the experiment does not claim to validate dataplane reachability or vendor CLI execution semantics that lie outside the verifier’s encoding.

Design implications for future NetOps evaluations. The paired `shared_initial_candidate` design cleanly isolates repair capability; however, operational questions remain that require alternative designs: (a) independent guarded-first-pass sampling would quantify whether guarded prompting or constrained generation yields a higher baseline of valid samples before repair; (b) multi-sample selection (n-best) could reveal selection-improvement effects; and (c) multi-model comparisons would quantify model-dependent failure patterns. From an experimental-planning perspective our results highlight that preregistered power calculations must account for plausible high baseline performance—if baseline rates turn out near 100% the discordant-pair approach will often be underpowered to detect small absolute repairs-only gains.

Reproducibility, artifact usage, and recommended reanalyses. All artifacts necessary to reproduce analysis are archived: `execution/task_manifest.jsonl` (deterministic task selection and `required_sequence` fields), `execution/model_configuration.json` (declared model, noting `temperature` = null/unset), `execution/responses/` (shared-initial and per-condition responses), `execution/scoring/` (per-episode

validator traces), `analysis/paired_contingency_table.csv` and `analysis/condition_summary.csv`, `deterministic_reconciliation.json` (provider accounting and `stage_counts`), and `analysis/analysis_log.jsonl` (bootstrap parameters and seed). Re-running the `paired_binary_analysis_v1` executor on the archived `input_results_sha256` reproduces the confirmatory results. For targeted reanalysis, researchers can (i) recompute bootstrap intervals under alternative resampling schemes, (ii) treat the single repaired task as an outlier to study robustness, or (iii) reclassify manifest invariants to evaluate sensitivity of verifier coverage—all without altering original model-call artifacts.

Threats to validity revisited (artifact-grounded). Internal validity: `shared_initial_candidate` semantics intentionally eliminate differences due to separate sampling, but they also prevent measuring effects of guarded prompt variants; model sampling nondeterminism remains possible because `temperature` was unset (`execution/model_configuration.json` records `temperature` = null). Construct validity: the verifier enforces only encoded invariants; omissions in the manifest (for example, dataplane-dependent invariants) are not tested. External validity: a single hosted model (gpt-5-mini) and a synthetic deterministic task bank limit generalizability; results do not imply identical outcomes for other models, live devices, or telemetry-driven validation. Conclusion validity: with one discordant pair the statistical evidence for a nonzero repair-only effect is weak ($p = 1.0$), though the bootstrap CI constrains plausible magnitudes.

Operational implications and cautious hypotheses. Artifact-grounded evidence demonstrates that bounded verifier-triggered repairs can fix localized ordering violations that are detectable and expressible in task manifests. From an operational standpoint this suggests a practical pattern: require explicit, machine-checkable manifest invariants for change tasks and pair LLM synthesis with deterministic validation/repair gates before human or automated deployment. We emphasize these are actionable hypotheses supported by the archived diagnostics in this controlled synthetic setting; they are not operational claims that bounded repair will resolve dataplane-level or adversarially induced failures without further evaluation.

Summary of expanded diagnostics. The archived execution bundle provides the evidentiary chain: deterministic task manifests encoding the invariants, the shared initial generated candidates, per-episode validator traces showing before/after states, provider-call accounting confirming `shared_initial_generation` = 200 and `repair` = 1, and analysis artifacts (contingency table, bootstrap parameters) documenting the statistical regime. Together these artifacts ground the narrower, repair-only conclusion and illuminate concrete design decisions for future, broader evaluations (independent guarded sampling, multi-model studies, live emulation).

VI. LIMITATIONS

- Synthetic deterministic task bank and verifier: results reflect correctness under the chosen synthetic intent tem-

plates and deterministic verifier and do not guarantee similar performance on live heterogeneous production devices or telemetry.

- Single-model experiment (gpt-5-mini): findings do not demonstrate multi-model generality.
- Shared_initial_candidate semantics: baseline and guarded conditions began from the same initial generation; the study measures verifier/repair effects only and does not evaluate independent first-pass generation contrasts.
- Limited adversarial/contamination testing: the run did not include systematic adversarial retrieval or poisoned-context attacks; such threats remain an open evaluation axis.
- Oracle coverage: the deterministic verifier enforces encoded invariants but may omit runtime behaviors and device-specific semantics observable only through live execution; external validity to live deployments is therefore limited.

VII. CONCLUSION

We executed a preregistered paired experiment measuring the marginal effect of a verifier-triggered repair under shared_initial_candidate semantics for LLM-generated network configurations. In 200 synthetic deterministic tasks with gpt-5-mini and deterministic_netops_validator_v5, baseline acceptance was 199/200 and guarded acceptance was 200/200; a single automated repair corrected a multi-device ordering violation. Paired absolute risk difference = 0.005 with paired-bootstrap 95% CI [0.00, 0.015] (10,000 resamples, seed 1729) and exact two-sided McNemar $p = 1.0$. Artifact-grounded evidence therefore supports a constrained conclusion: verifier-guarded bounded repair can correct localized ordering faults in this synthetic verifier-backed setting. Broader operational claims require additional experiments: multi-model studies, independent-first-pass guarded semantics, adversarial contamination testing, and live-device or dataplane emulation to capture runtime semantics not modeled by the deterministic verifier.

DISCLOSURE STATEMENT

Disclosure: This run implemented preregistered study CAND-2026-001 and was executed autonomously after the autonomy lock. Declared model: gpt-5-mini (execution/model_configuration.json records temperature = null/unset; null/unset is not evidence of deterministic sampling or temperature = 0). Orchestration adapter: hosted_netops_gvr_v1. Exact initial master prompt archived at provenance/master_prompt.txt (SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b b39b15ba). Preregistration fixed generation_semantics = shared_initial_candidate (one sampled initial generation per task was deterministically reused by baseline and guarded episodes) and allowed at most one automated repair call per task. Model-call accounting in the archived execution manifest reflects 200 shared initial generation calls and 1 repair call (201 model calls total). The deterministic verifier

used was deterministic_netops_validator_v5. Humans selected the broad topic family and constrained the initial master prompt prior to the autonomy lock as permitted by the track; no human manually edited technical manuscript content after the autonomy lock. All provider traces, verifier logs, task manifests, model-configuration records, and analysis artifacts are archived in the execution bundle and indexed by the execution manifest.

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